

Ahsan Mustafa

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SUMMARY

AI Developer with hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and full-stack web development. Skilled in building intelligent applications using Python, Django, PyTorch, TensorFlow, OpenCV, and modern AI frameworks. Experienced in developing real-time computer vision systems, CNN-based image classification models, object detection pipelines, NLP applications, and responsive AI-powered web platforms.

Recently focused on AI-powered accessibility solutions including a Bilingual Sign Language Recognition System (ASL & PSL), YOLO-based detection applications, transformer-based NLP models, and interactive AI web applications. Experienced in integrating Computer Vision, Deep Learning, and Generative AI technologies to develop practical real-world solutions.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, SQL

AI / ML: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, CNNs, Transfer Learning, Object Detection, Image Classification

Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, Scikit-learn, NumPy, Pandas, Hugging Face Transformers, nltk

Web Development: Django, HTML, CSS, JavaScript, PHP, MySQL, SQLite, Streamlit

Tools & Platforms: Git, GitHub, XAMPP, VS Code

PROJECTS

Bilingual Sign Language Recognition System (ASL & PSL) – In Progress

Final Year Project – AI, Computer Vision, Deep Learning

Python, OpenCV, TensorFlow/PyTorch, Gemini API

Developing a bilingual sign language recognition system supporting both American Sign Language (ASL) and Pakistan Sign Language (PSL).

Implementing real-time hand gesture recognition using Computer Vision and Deep Learning techniques.

Integrated Gemini API for intelligent sentence generation from recognized ASL words and phrases.

VocaSign –Alphabet Predictor

Deep Learning / Computer Vision

PyTorch, OpenCV, CNN, ResNet18

Developed a real-time sign language recognition system for alphabet-to-text/speech translation

Implemented CNN and transfer learning techniques using ResNet18 for gesture classification

Integrated real-time webcam processing using OpenCV for live predictions

AI Travel Planner

AI-Based Travel Recommendation System

Python, Streamlit, API Integration

Developed an AI-powered travel planner that generates personalized travel itineraries

Integrated APIs and user preference analysis for destination recommendations

Designed an interactive Streamlit-based interface for real-time travel planning

Spam Email Detector

Machine Learning / NLP

Python, Scikit-learn

Built a spam email classification system using machine learning and NLP preprocessing

Performed TF-IDF feature extraction, dataset analysis, and model evaluation

Sentiment Analysis using NLP

Natural Language Processing

Python, NLTK, Pandas

Developed a sentiment analysis pipeline to classify text sentiment using NLP techniques
Performed text preprocessing, sentiment scoring, and data analysis on textual datasets.

Word Embedding and TF-IDF Analysis

Natural Language Processing

Python, Scikit-learn, Hugging Face

Explored text vectorization techniques including TF-IDF and transformer embeddings
Implemented NLP workflows using pre-trained transformer models

Clothing Brand Database Management System

Full-Stack Web / Database

PHP, MySQL, CSS

Built a CRUD-based business management system for products, suppliers, customers, and sales
Designed SQL queries and analytical reports for business intelligence and sales tracking

Django Blog & Stopwatch Applications

Web Development

Python, Django, SQLite, HTML/CSS

Created a multi-user blogging platform with authentication, posting, and commenting functionality
Developed a stopwatch web application demonstrating frontend-backend interaction

EXPERIENCE

General Secretary, Arfa Karim Technology Incubator Society

Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

Led planning and coordination for technology-focused events and workshops
Collaborated with faculty and external partners to support innovation initiatives
Managed communications and operational activities across multiple domains

Security and Logistics Lead, IM—Hackathon

Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

Headed the security and logistics team for the university's first-ever hackathon
Coordinated infrastructure, scheduling, and operational planning
Worked with volunteers and administration for smooth event execution

Head of Events (Table Tennis), IM Olympiad

Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

Organized and managed table tennis events during the university's flagship sports competition
Handled scheduling, registrations, match fixtures, and logistics
Coordinated with sports committees and volunteers for smooth operations

Member, Proctorial Board

Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

Supported campus discipline, event coordination, and student management
Acted as liaison between administration and students during official events

EDUCATION

Institute of Management Sciences

Bachelor of Science in Software Engineering

*Hayatabad, Peshawar
2022 – 2026*

Edwards College

A-Levels

*Peshawar
2018 – 2020*

Edwards College School

O-Levels

*Peshawar
2016 – 2018*

CERTIFICATIONS

Python and Django Full Stack Web Developer Bootcamp

Complete Computer Vision Bootcamp with PyTorch & TensorFlow

Generative AI Course — Arfa Karim Technology Incubator Peshawar

Two-Day Agentic AI Workshop during National AI Hackathon 2026, IMSciences Peshawar (AtomCamp)

Python Data Structures

Basics of Cloud Computing — Fundamentals of Cloud Computing